

First order and first degree differential equation:

Any first order & first degree diff. eq. can be written as.

$$M(x,y) \frac{dy}{dx} + N(x,y) = 0,$$

or equivalently, $\frac{dy}{dx} = f(x,y)$, when M , N and f are functions of x, y .
— (1)

Case-1 Variable separable Method.

If $\frac{dy}{dx} = f_1(x) \cdot f_2(y)$, then

$$\frac{dy}{f_2(y)} = f_1(x) dx$$

On integrating, $\boxed{\int \frac{dy}{f_2(y)} = \int f_1(x) dx + C}$

Ex-1: $\frac{dy}{dx} = -2xy$.

Sol: $\frac{dy}{y} = -2x dx$

On integrating $\ln y = -x^2 + C$, C - constant.

$$y = e^{-x^2+C} = c_1 e^{-x^2}, \quad c_1 = e^C.$$

Ex-2 $y' = -ky$, k - constant

Sol $\frac{dy}{y} = -k dx \Rightarrow \ln y = -kx + C$

$$y = e^{-kx+C} = c_1 e^{-kx}, \quad c_1 = e^C$$

Ex-3 $y' = \frac{y \cos x}{1+2y^2}$, $y(0) = 1$.

Sol $\frac{1+2y^2}{y} dy = \cos x dx$

on integrating, $\int \frac{1}{y} + 2y dy = \int \cos x dx + C$

$\Rightarrow \ln y + y^2 = \sin x + C$

Put $x=0$, $\ln y(0) + (y(0))^2 = \sin 0 + C \Rightarrow \ln 1 + 1 = C$
 $\Rightarrow C=1 \quad \therefore \boxed{\ln y + y^2 = \sin x}$

Ex.4. $y' = \frac{1+y^2}{1+x^2}$

Ex.5. $y' = x \sin y$

Ex.6 $y' = x^2 \cos y$

Ex.7. $y' = y \cot x$, $y(\pi/2) = 1$ Ex.8 $(x^2+1)dy + (y^2+4)dx = 0$

Homogeneous differential Eq. (Reducible to variable separable):

Homogeneous function: A function $f(x, y)$ is said to be homogeneous of degree n if

$f(x, vx) = x^n \phi(v)$, ϕ is a function of v .

Ex $f(x, y) = x + y$
 $f(x, vx) = x + vx = x(1+v)$ (homogeneous of degree 1).

Ex $f(x, y) = x^2 - y^2$
 $f(x, vx) = x^2 - v^2x^2 = x^2(1-v^2)$ (homogeneous of degree 2).

Ex $f(x, y) = e^{x/y} + \cos y/x$
 $f(x, vx) = e^{x/vx} + \cos \frac{vx}{x} = e^{1/v} + \cos v$
 (homogeneous of degree 0).

Suppose $\frac{dy}{dx} = f(x, y) = \frac{f_1(x, y)}{f_2(x, y)}$ Eq (2)

If f_1 and f_2 are homogeneous functions of the same degree n then diff eq (2) can be reduced to variable separable form by taking $y = vx$, v is a function of x .

$$\text{Let } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore v + x \frac{dv}{dx} = \frac{f_1(x, vx)}{f_2(x, vx)} = \frac{x^n \phi_1(v)}{x^n \phi_2(v)}$$

$$x \frac{dv}{dx} = \frac{\phi_1(v)}{\phi_2(v)} - v \quad (\text{it is in the variable separable form}).$$

Ex.1 $\frac{dy}{dx} = \frac{x+y}{x-y}$

Solⁿ: Both numerator and denominator are homogeneous of degree 1. Take $y = vx$.

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore v + x \frac{dv}{dx} = \frac{x+vx}{x-vx} = \frac{1+v}{1-v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v}{1-v} - v = \frac{1+v-v+v^2}{1-v} = \frac{1+v^2}{1-v}$$

$$\Rightarrow \frac{1-v}{1+v^2} dv = \frac{dx}{x}$$

On integrating $\int \frac{1}{1+v^2} - \frac{v}{1+v^2} dv = \int \frac{dx}{x} + C$

$$\Rightarrow \tan^{-1}v - \frac{1}{2} \ln(1+v^2) = \ln x + C$$

$$\Rightarrow \tan^{-1} \frac{y}{x} - \frac{1}{2} \ln \left(1 + \frac{y^2}{x^2} \right) = \ln x + C.$$

Ex.2 $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$

Ex.3 $y' = \frac{y^2 - xy}{x^2 + xy}$

Ex.4 $y' = \frac{y^2 + xy + x^2}{x^2}$

Ex.5 $y' = \frac{y + x \cos^2(y/x)}{x}$

Reducible to homogeneous diff. Eq.

$$\frac{dy}{dx} = \frac{ax + by + m}{cx + dy + n}, \quad a, b, c, d, m, n - \text{constants.}$$

It can be reduced to homogeneous diff eq if $ad-bc \neq 0$.

Put $x = X+h$ $y = Y+k$

$$\frac{dy}{dx} = \frac{dY}{dX}$$

$$\therefore \frac{dY}{dX} = \frac{a(X+h) + b(Y+k) + m}{c(X+h) + d(Y+k) + n}$$

Assume $ah + bk + m = 0 \Rightarrow ah + bk = -m$
 $ch + dk + n = 0 \Rightarrow ch + dk = -n$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} h \\ k \end{bmatrix} = - \begin{bmatrix} m \\ n \end{bmatrix}$$

It has a unique solⁿ if $ad-bc \neq 0$.

$$\therefore \frac{dY}{dX} = \frac{aX + bY}{cX + dY}$$

Ex. $\frac{dy}{dx} = - \frac{1+x-2y}{4x-3y-6}$

Sol: Let $x = X+h$ and $y = Y+k$.

$\therefore$ The above diff. eq. reduces to

$$\frac{dY}{dX} = - \frac{1+X+h-2(Y+k)}{4(X+h)-3(Y+k)-6} = - \frac{X-2Y+1+h-2k}{4X-3Y-6+4h-3k}$$

Set $\begin{cases} 1+h-2k=0 \Rightarrow h-2k=-1 \\ -6+4h-3k=0 \Rightarrow 4h-3k=6 \end{cases} \Rightarrow h=3, k=2.$

$$\therefore \frac{dY}{dX} = - \frac{X-2Y}{4X-3Y} \quad (\text{This is a homogeneous diff. eq.})$$

$$\therefore \text{let } Y = VX \Rightarrow \frac{dY}{dX} = V + X \frac{dV}{dX}$$

$$\therefore V + X \frac{dV}{dX} = - \frac{X-2VX}{4X-3VX} = - \frac{1-2V}{4-3V}$$

$$\begin{aligned} \Rightarrow X \frac{dV}{dX} &= - \frac{1-2V}{4-3V} - V = \frac{-1+2V-4V+3V^2}{4-3V} \\ &= \frac{-1-2V+3V^2}{4-3V} = \frac{(3V+1)(V-1)}{4-3V} \end{aligned}$$

$$\Rightarrow \frac{4-3v}{(3v+1)(v-1)} dv = \frac{dx}{x}$$

$$\Rightarrow \left[-\frac{15}{4(3v+1)} + \frac{1}{4(v-1)} \right] dv = \frac{dx}{x}$$

On integrating, $-\frac{15}{4} \frac{\ln(3v+1)}{3} + \frac{1}{4} \ln(v-1) = \ln x + \ln c.$

$$\Rightarrow -\frac{5}{4} \ln(3v+1) + \ln(v-1)^{1/4} = \ln x + \ln c$$

$$\Rightarrow \ln(3v+1)^{-5/4} (v-1)^{1/4} = \ln xc$$

$$\Rightarrow (3v+1)^{-5/4} (v-1)^{1/4} = xc \Rightarrow (v-1)^{1/4} = xc(3v+1)^{5/4}$$

Put $v = y/x$, $\frac{(y-x)^{1/4}}{x^{5/4}} = xc \frac{(3y+x)^{5/4}}{x^{5/4}}$

$$\Rightarrow (y-x)^{1/4} = c(3y+x)^{5/4}$$

Put $x = x-h = x-3$ and $y = y-k = y-2$.

$$\therefore (y-2-(x-3))^{1/4} = c(3(y-2)+x-3)^{5/4}$$

$$\therefore \boxed{(y-x+1)^{1/4} = c(3y+x-9)^{5/4}}$$

Ex. $\frac{dy}{dx} = \frac{2y-x-4}{y-3x+3}$

Ex $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$

Ex. $\frac{dy}{dx} = \frac{x+y+3}{2x+2y+1}$

(Hint: Put $x+y=v$)

Ex $\frac{dy}{dx} = \frac{x+2y-1}{x+2y+1}$

(Hint: Put $x+2y=v$)